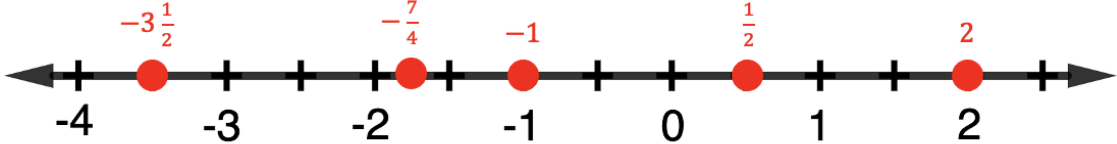
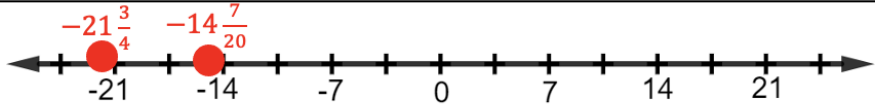


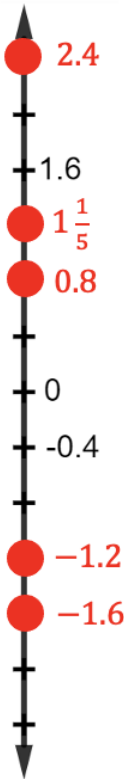
6th Grade Unit 4 Assessment Guide

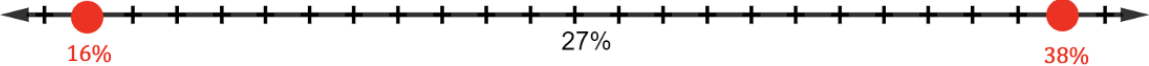
General Information	The benchmarks assessed on this assessment focus on number sense of rational numbers. Students learn the importance of a number's distance from zero. This knowledge is utilized when comparing and ordering negative values as well as in understanding absolute value. The foundation of absolute value being the distance from zero is important for student understanding of algebraic absolute value equations. Avoid teaching students shortcuts such as "just make it positive" which can confuse students later as they study algebraic absolute value equations. As seen throughout the assessment, MA.6.NSO.1.1 can be assessed in many different ways. Growing student flexibility in applying their understanding of this benchmark will also help increase a student's number sense. Frequency of a student's interaction with rational numbers can support their fluency and number sense. Teachers are encouraged to present students with a variety of rational numbers in various forms throughout the remainder of the school year.
Calculator Information	The assessment is calculator neutral. The benchmarks on this assessment are calculator neutral which means that a calculator may be used for some items on the statewide assessment. Each teacher should determine if a calculator is appropriate for their individual classroom. The benchmarks do not state that students have to show knowledge of converting numbers to different forms but that their knowledge is plotting, ordering, and comparing. If teachers do not wish for their students to use a calculator, then the length of some questions may need to be modified.
Total Recommended Points	The total recommended points in this guide is 100 points.

Question	1a	Benchmark(s)	MA.6.NSO.1.2
	The value represented by point ☐ B ☐ H ☐ K ☐ T ☐ W is the opposite of the value represented by point R because point R is the same distance from ☐ -1 ☒ 0 ☐ 1 as point ☒ B. ☐ H. ☐ K. ☐ T. ☐ W. 	Recommended Scoring (4 Total Points)	No partial credit is suggested. Students should correctly answer all three answer boxes to show knowledge of the benchmark.
Question	1b	Benchmark(s)	MA.6.NSO.1.1
	☒ B ☒ H ☐ K ☐ R ☒ W	Recommended Scoring (6 Total Points)	Consider giving students 2 points for each point that is correctly chosen and deduct 3 points for each point that is incorrectly chosen.

Question	2	Benchmark(s)	MA.6.NSO.1.1
		Recommended Scoring (7 Total Points)	Consider giving students 3 points for the correct order of the negative values and 4 points for the correct order of the positive values.
Least to Greatest	$-3.1, -\frac{9}{4}, -2\frac{3}{20}, \frac{7}{8}, 89\%, 2\frac{1}{5}, 2.25$	Additional Information	If a teacher chooses not to allow their students to use a calculator, then it is suggested that this question be modified to reduce the cognitive demand on the student. It is suggested that the values $2\frac{1}{5}$ and 2.25 be removed in this case.
Question	3	Benchmark(s)	MA.6.NSO.1.1
		Recommended Scoring (10 Total Points)	Consider giving students 2 points for each correctly plotted point.

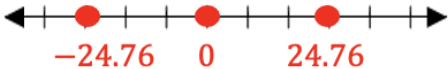
Question		4	Benchmark(s)	MA.6.NSO.1.1
Number Line			Recommended Scoring (6 Total Points)	Consider giving students 2 points for the correct number line and 4 points for the correct explanation.
Explanation	Since both $-14\frac{7}{20}$ and $-21\frac{3}{4}$ are negative numbers, the larger value will be the value closer to 0. $-14\frac{7}{20}$ is closer to zero, so Maggie's statement is not correct.			

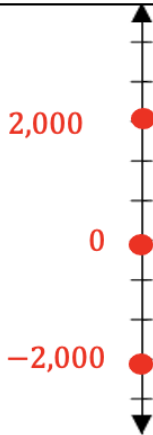
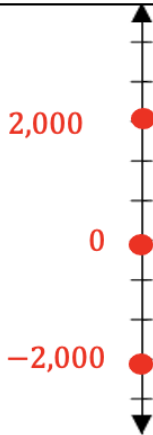
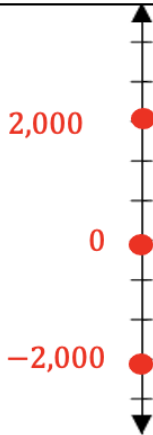
Question		5	Benchmark(s)	MA.6.NSO.1.1
		Recommended Scoring (10 Total Points)	Consider giving students 2 points for each correctly plotted point.	

Question	6a	Benchmark(s)	MA.6.NSO.1.3
		Recommended Scoring (4 Total Points)	No partial credit is suggested. To show knowledge, students should plot 16% and 38% so that each is the same distance from 27%. The number line allows student to use 1 tick mark for 1%, but students can use any number of tick marks as long the number of tick marks between 27% and the other two points are the same.
Question	6b	Benchmark(s)	MA.6.NSO.1.3
The 11% difference between the 2005 and 2021 survey represented by absolute value because the two possible results of the 2021 survey	☐ can ☐ cannot be ☐ are ☐ are not the same distance from ☐ zero. ☐ the zero point, 27%.	Recommended Scoring (3 Total Points)	No partial credit is suggested. Students should correctly answer all three answer boxes to show knowledge of the benchmark.

Question	7a	Benchmark(s)	MA.6.NSO.1.1
mathematical statement $-226.35 < -63.72$		Recommended Scoring (3 Total Points)	No partial credit is suggested. Students may write the inequality as $-63.72 > -226.35$.
Question	7b	Benchmark(s)	MA.6.NSO.1.4
mathematical statement $-226.35 + -63.72$		Recommended Scoring (3 Total Points)	No partial credit is suggested. Students may write the statement as $ -63.72 + -226.35 $.
Question	7c	Benchmark(s)	MA.6.NSO.1.4
Henry's Total Debt $\\$290.07$		Recommended Scoring (3 Total Points)	No partial credit is suggested.

Question	8a	Benchmark(s)	MA.6.NSO.1.4
$ -65 > -63 $		Recommended Scoring (3 Total Points)	No partial credit is suggested.
Question	8b	Benchmark(s)	MA.6.NSO.1.1
$83.5\% < \frac{7}{8}$		Recommended Scoring (3 Total Points)	No partial credit is suggested.
Question	8c	Benchmark(s)	MA.6.NSO.1.1
$-12.5 = -\frac{225}{18}$		Recommended Scoring (3 Total Points)	No partial credit is suggested.
Question	8d	Benchmark(s)	MA.6.NSO.1.4
$14 < -23 $		Recommended Scoring (3 Total Points)	No partial credit is suggested.

Question	9 (A withdrawal of \$24.76)		Benchmark(s)	MA.6.NSO.1.2
		A withdrawal of \$24.76	Recommended Scoring (10 Total Points)	Consider giving students points as follows: • 2 points for the correct rational number • 2 points for the correct opposite • 3 points for the correct number line that shows the rational number and its opposite in relation to 0. • 3 points for the correct interpretation of zero.
	Rational Number	-24.76		
	Opposite	24.76		
	Number Line			
	Meaning of Zero	Zero represents the initial amount of money in the account.		

Question	9 (2000 meters above sea level)	Benchmark(s)	MA.6.NSO.1.2										
	<table><tr><td></td><td>2000 meters above sea level</td></tr><tr><td>Rational Number</td><td>2,000</td></tr><tr><td>Opposite</td><td>-2,000</td></tr><tr><td>Number Line</td><td></td></tr><tr><td>Meaning of Zero</td><td>Zero represents sea level.</td></tr></table>		2000 meters above sea level	Rational Number	2,000	Opposite	-2,000	Number Line		Meaning of Zero	Zero represents sea level.	Recommended Scoring (10 Total Points)	Consider giving students points as follows: • 2 points for the correct rational number • 2 points for the correct opposite • 3 points for the correct number line that shows the rational number and its opposite in relation to 0. • 3 points for the correct interpretation of zero.
	2000 meters above sea level												
Rational Number	2,000												
Opposite	-2,000												
Number Line													
Meaning of Zero	Zero represents sea level.												

Question	10a	Benchmark(s)	MA.6.NSO.1.4
value 68		Recommended Scoring (3 Total Points)	No partial credit is suggested.
Question	10b	Benchmark(s)	MA.6.NSO.1.4
value 31		Recommended Scoring (3 Total Points)	No partial credit is suggested.
Question	10c	Benchmark(s)	MA.6.NSO.1.4
value 106		Recommended Scoring (3 Total Points)	No partial credit is suggested.